

1 Multiplicative differential germs

Source: PrimeTensor/Analysis/Derivative.lean

What this file establishes

This is where the development starts doing calculus, and the module header is emphatic about how: the multiplicative derivative is *not* obtained by translating the additive difference quotient. Four notions are introduced.

- A *perturbation* is a sequence of completed magnitudes tending to the pivot 1. It replaces “ $h \rightarrow 0$ ”.
- The *response* of f at x along h is the sequence of output ratios $f(x \cdot h_n)/f(x)$. It replaces the difference $f(x + h) - f(x)$ — and note that nothing is divided by the perturbation.
- A *tangent map* is a morphism of the multiplicative carrier. It replaces the linear map.
- *Negligibility* of one germ relative to another is stated by demanding that the first reach a scale level an arbitrary prescribed number of refinements deeper than the second. It replaces little-o.

Assembling these, D is a derivative of f at x when it models every pivot-approaching perturbation to first order. The file closes by computing the two derivatives that can be computed with no analysis at all: the identity function has the identity tangent map, and every constant function has the trivial one.

Design note 1.9 works out what all four notions say in the logarithmic chart, which is the quickest way to see that they are the familiar ones and not merely analogous to them.

1.1 Advancing a scale

Definition 1.1 (`Depth.advance`). `advance : Depth → Depth → Depth` is defined by recursion on its second argument:

$$\text{advance}(\ell, \text{one}) = \text{succ } \ell, \quad \text{advance}(\ell, \text{succ } g) = \text{succ } \text{advance}(\ell, g).$$

```
def advance : Depth → Depth → Depth
| level, .one => .succ level
| level, .succ gain => .succ (advance level gain)
```

So $|\text{advance}(\ell, g)| = |\ell| + |g|$: the level is refined $|g|$ times. As the docstring says, there is deliberately no advance by zero — `one` already means one refinement. A consequence worth recording now is that the vocabulary can express “strictly finer by a prescribed amount” and cannot express “no finer”: there is no way to write the multiplicative analogue of a big- O relation in this file, only of a little-o one.

1.2 Ratios in the completion

Definition 1.2 (`MulReal.ratio`). $\text{ratio}(a, b) \equiv a \cdot b^{-1}$, the completed counterpart of the ratio of `PrimeTensor/Order.lean`.

Lemma 1.3 (`MulReal.ratio_self`, `inv_one` and `ratio_mul_base`). $\text{ratio}(a, a) = 1$; $1^{-1} = 1$; and for all x, h ,

$$\text{ratio}(x \cdot h, x) = h.$$

```
theorem ratio_mul_base (x h : MulReal) :
  ratio (x * h) x = h := by
  unfold ratio
  calc
    (x * h) * x⁻¹
      = h * (x * x⁻¹) := by
        rw [mul_comm x h]
        exact mul_assoc h x x⁻¹
    _ = h * 1 := by rw [mul_inv x]
    _ = h := mul_one h
```

Proof. The first is `mul_inv` of `PrimeTensor/Algebra.lean`. For the second, $1 \cdot 1^{-1} = 1$ becomes $1^{-1} = 1$ after cancelling the unit on the left. For the third, commute x past h , reassociate to expose xx^{-1} , cancel it, and drop the unit. $\square$

The third identity is small but it is the pivot of the whole file: it says that perturbing x multiplicatively by h and then comparing with x returns exactly h . The response of the identity function is therefore the perturbation itself, on the nose, with no error term — which is Theorem 1.13 below.

1.3 Tangent maps

Definition 1.4 (`MulTangentMap`). A *tangent map* is a structure bundling a function `toFun : MulReal → MulReal` with proofs that it preserves the pivot, the multiplication and the inversion. A `CoeFun` instance lets such a structure be applied directly to an argument.

```
structure MulTangentMap where
  toFun : MulReal → MulReal
  map_one : toFun 1 = 1
  map_mul : ∀ a b : MulReal, toFun (a * b) = toFun a * toFun b
  map_inv : ∀ a : MulReal, toFun a⁻¹ = (toFun a)⁻¹
```

Remark 1.5 (This is the linear map, and the axiom list is again redundant). In the additive theory the first-order model is required to be linear. Here the requirement is that the model be a morphism of the structure the carrier actually has — the one axiomatised as `IsMulCarrier` in `PrimeTensor/Structure.lean` — and multiplicativity is what additivity becomes.

As with that class, the axioms are not independent: `map_inv` follows from the other two. From $\text{toFun}(a^{-1})\text{toFun}(a) = \text{toFun}(a^{-1}a) = \text{toFun}(1) = 1$ and uniqueness of inverses in a carrier, $\text{toFun}(a^{-1})$ is $\text{toFun}(a)^{-1}$. It is listed because it is used, and listing it costs one line in each of the two instances below.

What is *not* required is any continuity or boundedness of the tangent map. In the additive theory over an infinite-dimensional space that requirement is what makes the derivative well

behaved; here nothing rules out a wildly discontinuous morphism serving as a first-order model, and nothing in this file needs to.

Definition 1.6 (`MulTangentMap.identity` and `trivial`). `identity` is $h \mapsto h$ and `trivial` is $h \mapsto 1$; the `simp` lemmas `identity_apply` and `trivial_apply` record their values, both by `rfl`.

Verification. For `identity` all three laws are `rfl`. For `trivial`, preservation of the pivot is `rfl`, preservation of the product is $1 = 1 \cdot 1$ read backwards from `one_mul`, and preservation of the inverse is $1 = 1^{-1}$ read backwards from Lemma 1.3. $\square$

1.4 Perturbations, responses and negligibility

Definition 1.7 (`ApproachesPivot` and `response`). A sequence $h : \text{Depth} \rightarrow \text{MulReal}$ *approaches the pivot* when it converges to 1 in the sense of `PrimeTensor/Analysis/Limit.lean`. The *response* of f at x along h is the sequence

$$\text{response}(f, x, h)_n \equiv \text{ratio}(f(x \cdot h_n), f(x)).$$

Two features distinguish this from a difference quotient. The perturbation enters multiplicatively, as $x \cdot h_n$ rather than $x + h_n$, so the statement never needs an additive structure on the domain. And the response is *not* divided by the perturbation: it is the raw output ratio, and the comparison against the model happens in Definition 1.10 through negligibility rather than through a quotient. This matters because dividing would require the model to be invertible, which no tangent map is asked to be — `trivial` is not.

Definition 1.8 (`NegligibleRelative`). `error` is *negligible relative to base* when

$$\forall g \exists c \forall n \text{ at or after } c \forall \ell : \text{ScaleNear}(\ell, \text{base}_n, 1) \rightarrow \text{ScaleNear}(\text{advance}(\ell, g), \text{error}_n, 1).$$

```
def NegligibleRelative (error base : MulReal.Seq) : Prop :=
  ∀ gain : Depth,
    ∃ anchor : Depth,
      ∀ n : Depth,
        Depth.AtOrAfter anchor n →
          ∀ level : Depth,
            MulReal.ScaleNear level (base n) 1 →
              MulReal.ScaleNear (Depth.advance level gain) (error n) 1
```

Read it as a scale shift: whatever level the base achieves at stage n , the error achieves that level advanced by g , and g may be prescribed in advance and arbitrarily large. Two points of quantifier order are worth noticing. The anchor is chosen before the level, so one anchor must serve for every level simultaneously — the shift is uniform in ℓ . And the implication is per-stage, so at a stage where the base achieves no level at all the condition says nothing about the error there; the definition constrains the error only where the base is genuinely small, which for a perturbation is eventually everywhere.

Design note 1.9 (All of this in the logarithmic chart). Write $L = \log_2$, so that L carries the multiplicative carrier to an additive one, $L(1) = 0$, and $L(\text{ratio}(a, b)) = L(a) - L(b)$. By the computation of `PrimeTensor/Scale.lean`, closeness at level ℓ is

$$\text{ScaleWithin}(\ell, a, b) \iff |L(a) - L(b)| < 2^{-(|\ell|-1)},$$

so a level is a dyadic tolerance and `succ` halves it. Under L :

Perturbations approaching the pivot become sequences with $L(h_n) \rightarrow 0$. *Responses* become $Lf(x \cdot h_n) - Lf(x)$, that is, the increment of $L \circ f$ at the argument increment $L(h_n)$. *Tangent maps* become additive maps. And *negligibility*, since $\text{advance}(\ell, g)$ adds $|g|$ to the level, becomes

$$|L(\text{base}_n)| < 2^{-(|\ell|-1)} \implies |L(\text{error}_n)| < 2^{-|g|} \cdot 2^{-(|\ell|-1)},$$

which is $|L(\text{error}_n)| \leq 2^{-|g|}|L(\text{base}_n)|$ up to the factor of at most two lost by rounding the bound to the next dyadic value. Since g is arbitrary, that is exactly $\text{error} = o(\text{base})$, with the rate quantified in powers of two rather than by an arbitrary positive real.

Putting the four together, Definition 1.10 says

$$Lf(x \cdot h_n) - Lf(x) = \tilde{D}(L(h_n)) + o(L(h_n)), \quad \tilde{D} \text{ additive},$$

which is the ordinary first-order expansion of $L \circ f \circ L^{-1}$ at $L(x)$. The multiplicative derivative of f is the ordinary derivative of the conjugated function; the content of the file is that this can be said without ever leaving the multiplicative language, so that no logarithm, no real exponent and no additive ε is introduced. The chart is a reading aid, and nothing in this paragraph is proved in Lean.

Definition 1.10 (`FirstOrderEquivalent` and `HasMulDerivativeAt`). Germs `actual` and `model` agree to first order relative to `base` when $n \mapsto \text{ratio}(\text{actual}_n, \text{model}_n)$ is negligible relative to `base`. And D is a multiplicative derivative of f at x when for every perturbation h approaching the pivot, the response of f at x along h agrees to first order, relative to h , with $n \mapsto D(h_n)$.

```
def HasMulDerivativeAt
  (f : MulReal → MulReal) (x : MulReal) (D : MulTangentMap) : Prop :=
  ∀ h : MulReal.Seq,
    ApproachesPivot h →
      FirstOrderEquivalent h
        (response f x h)
        (fun n => D (h n))
```

The error between the actual response and the model is measured as their *ratio*, which is the multiplicative counterpart of their difference; the two are the same object in the chart of Design note 1.9.

Remark 1.11 (Sequential, not filtered). The definition quantifies over sequences h rather than over a neighbourhood filter of the pivot, so it is a sequential notion — closer to a Gateaux derivative along all pivot-approaching sequences than to a Fréchet derivative. Nothing here asserts that the scale shift is uniform over the choice of h , only over the level for each fixed h .

1.5 The two computable cases

Lemma 1.12 (`firstOrder_of_exact`). If $\text{actual}_n = \text{model}_n$ for every n , then they agree to first order relative to any `base`.

Proof. Given a gain, take the anchor `one`. For any n and any level, the ratio of the two germs at n is $\text{ratio}(m, m) = 1$ by Lemma 1.3, and $\text{ScaleNear}(\ell', 1, 1)$ holds at every level ℓ' by reflexivity. The hypothesis about the base is not used. $\square$

Theorem 1.13 (`hasMulDerivativeAt_identity`). *The identity function has derivative identity at every point.*

Proof. Fix x and a pivot-approaching h . The response at stage n is $\text{ratio}(x \cdot h_n, x)$, which is h_n by Lemma 1.3, and the model at stage n is $\text{identity}(h_n) = h_n$. The two germs are equal term by term, so Lemma 1.12 applies. $\square$

Theorem 1.14 (`hasMulDerivativeAt_constant`). *For every c , the constant function at c has derivative trivial at every point.*

Proof. The response at stage n is $\text{ratio}(c, c) = 1$ and the model is $\text{trivial}(h_n) = 1$; again the germs are equal term by term. $\square$

Remark 1.15 (Neither example exercises the machinery). Both theorems go through Lemma 1.12, whose proof discards the base hypothesis and returns the anchor `one`. So neither computation uses `advance`, `negligibility`, the convergence of h , or any property of the scale beyond reflexivity of `ScaleNear`: the hypothesis `ApproachesPivot h` is accepted and never read in either proof. These are the two cases where the model is exact, and they are here as sanity checks on the definitions rather than as evidence that the definitions have content. The first genuinely approximate derivative is not computed in this file.

Remark 1.16 (What is not proved). The derivative is not shown unique: nothing rules out two different tangent maps both modelling f at x , and the argument that would exclude it needs the separation property still missing since `PrimeTensor/Structure.lean`. There is no chain rule, no product rule, and no statement that a function with a derivative is in any sense continuous. `NegligibleRelative` is not shown transitive, nor closed under products, nor related to `ConvergesTo`. And, consistently with the header, nothing additive is used anywhere: no subtraction, no zero, no zeroth index, and no additive ε .

Summary of the correspondence

Lean declaration	Kind	Here
<code>Depth.advance</code>	definition	Def. 1.1
<code>MulReal.ratio</code>	definition	Def. 1.2
<code>MulReal.ratio_self</code>	simp thm	Lem. 1.3
<code>MulReal.inv_one</code>	simp thm	Lem. 1.3
<code>MulReal.ratio_mul_base</code>	theorem	Lem. 1.3
<code>MulTangentMap</code>	structure	Def. 1.4
<code>CoeFun MulTangentMap</code>	instance	Def. 1.4
<code>MulTangentMap.identity</code>	definition	Def. 1.6
<code>MulTangentMap.trivial</code>	definition	Def. 1.6
<code>MulTangentMap.identity_apply</code>	simp thm	Def. 1.6
<code>MulTangentMap.trivial_apply</code>	simp thm	Def. 1.6
<code>MulDifferential.ApproachesPivot</code>	definition	Def. 1.7
<code>MulDifferential.response</code>	definition	Def. 1.7
<code>MulDifferential.NegligibleRelative</code>	definition	Def. 1.8
<code>MulDifferential.FirstOrderEquivalent</code>	definition	Def. 1.10
<code>MulDifferential.HasMulDerivativeAt</code>	definition	Def. 1.10
<code>MulDifferential.firstOrder_of_exact</code>	theorem	Lem. 1.12
<code>hasMulDerivativeAt_identity</code>	theorem	Thm. 1.13
<code>hasMulDerivativeAt_constant</code>	theorem	Thm. 1.14

Every declaration of `PrimeTensor/Analysis/Derivative.lean` appears above, together with the four fields of `MulTangentMap`. The file contains no `sorry`, no axiom, and no unproved named proposition. Design note 1.9 and Remarks 1.5, 1.11, 1.15 and 1.16 are stated for the reader and are not declarations of the file; in particular the logarithmic chart is a reading aid and is not formalised.